

Speed Symmetry and Two-Sided Exit Laws for Skew Brownian Motion

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Abstract

For zero-drift skew Brownian motion with symmetric local time we close the Lebesgue kernel, speed-symmetric semigroup, resolvent, and every two-sided exit probability, discounted side transform, and mean exit time. The exact formulas retain unequal interval endpoints and all skew chambers.

1 Kernel and the reference measure

Freeze $X_t = x + B_t + (2p - 1)L_t^0(X)$, $0 \leq p \leq 1$, with symmetric semimartingale local time [1]. With $\phi_t(z) = (2\pi t)^{-1/2} e^{-z^2/(2t)}$, its Lebesgue density is

$$q_p(t; x, y) = \phi_t(y - x) + (2p - 1) \operatorname{sgn}(y) \phi_t(|x| + |y|). \quad (1)$$

The generator is $f''/2$ off zero with $pf'(0+) = (1 - p)f'(0-)$. Gaussian convolution split at zero proves the semigroup law.

For $0 < p < 1$, set

$$m'(y) = \begin{cases} 2(1 - p), & y < 0, \\ 2p, & y > 0. \end{cases}$$

The Lebesgue density jumps in y at zero, but $k_p(t; x, y) = q_p(t; x, y)/m'(y)$ is symmetric. Indeed the two cross-interface Lebesgue densities are respectively $2(1 - p)\phi_t(|x| + |y|)$ and $2p\phi_t(|x| + |y|)$. Thus the Markov semigroup is self-adjoint on $L^2(m)$. For $\lambda > 0$, $k = \sqrt{2\lambda}$, its Lebesgue resolvent is

$$r_\lambda(x, y) = \frac{e^{-k|x-y|} + (2p - 1) \operatorname{sgn}(y) e^{-k(|x|+|y|)}}{k}. \quad (2)$$

2 Complete exit atlas

Let $\tau = \inf\{t : X_t \notin (-a, b)\}$, where $a, b > 0$. Use the scale derivatives $s' = 1/(1 - p)$ left of zero and $s' = 1/p$ right of zero.

Theorem 1 (undiscounted exits). *With $D = pa + (1 - p)b$,*

$$\Pr_x(X_\tau = b) = \begin{cases} p(x + a)/D, & -a \leq x \leq 0, \\ [pa + (1 - p)x]/D, & 0 \leq x \leq b. \end{cases}$$

Put $D_m = (1 - p)b + pa$ and

$$A = \frac{(1 - p)(b^2 - a^2)}{D_m}, \quad C = \frac{p(b^2 - a^2)}{D_m}, \quad B = \frac{ab[pb + (1 - p)a]}{D_m}.$$

Then

$$\mathbb{E}_x \tau = \begin{cases} -x^2 + Cx + B, & x \leq 0, \\ -x^2 + Ax + B, & x \geq 0. \end{cases}$$

Proof. Equation (3) is the unique piecewise affine solution of $h'' = 0$ with boundary values zero and one, continuity, and the skew derivative condition. Equation (4) is the unique piecewise quadratic solution of $m''/2 = -1$ with zero boundary values and the same two interface conditions. $\square$

Theorem 2 (discounted side transforms). *For $0 < p < 1$, set $\rho = (1 - p)/p$, $k = \sqrt{2\lambda}$, and*

$$Q = \cosh(kb) \sinh(ka) + \rho \sinh(kb) \cosh(ka).$$

Then $R_\lambda(x) = \mathbb{E}_x[e^{-\lambda\tau}; X_\tau = b]$ equals

$$\begin{cases} \sinh(k(x+a))/Q, & x \leq 0, \\ [\cosh(kx) \sinh(ka) + \rho \sinh(kx) \cosh(ka)]/Q, & x \geq 0. \end{cases}$$

The left transform is $R_\lambda^{1-p}(-x; b, a)$, and their sum is $\mathbb{E}_x e^{-\lambda\tau}$.

Proof. On both sides solve $u''/2 = \lambda u$. Propagation of $(u, u'/k)$ uses the hyperbolic matrix $\begin{pmatrix} \cosh(k\ell) & \sinh(k\ell) \\ \sinh(k\ell) & \cosh(k\ell) \end{pmatrix}$. At zero continuity retains the first coordinate while (3)'s derivative condition multiplies the second by ρ . Imposing $u(-a) = 0, u(b) = 1$ gives (5); spatial reflection gives the left side. $\square$

At $p = 1/2$ this is Brownian motion. The limits $p = 0, 1$ give one-sided reflection after hitting zero. No target arithmetic or determinant is used.

References

- [1] J. M. Harrison and L. A. Shepp, *On skew Brownian motion*, Ann. Probab. **9** (1981), 309–313, [doi:10.1214/aop/1176994472](https://doi.org/10.1214/aop/1176994472).